The documentation and process conversion measures necessary to comply with this document shall be completed by 1 May 2006.

INCH-POUND

MIL-PRF-19500/368H
1 February 2006
SUPERSEDING
MIL-PRF-19500/368G
1 May 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES: 2N3439, 2N3439L, 2N3439UA, 2N3439U4, 2N3440, 2N3440L, 2N3440UA,
AND 2N3440U4, JAN, JANTX, JANTXV, JANS, JANHCA, JANKCA, JANHCB, AND JANKCB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power, high voltage transistors. Four levels of product assurance are provided for each encapsulated device types as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figure 1 (similar to TO-5 and TO-39), figure 2 (JANHCA and JANKCA (A versions)), figure 3 (JANHCB and JANKCB (B versions)), figure 4 (2N3439UA and 2N3440UA surface mount versions), and figure 5 (2N3439U4 and 2N3440U4) versions).

* 1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Types | $P_T$ (1) $T_A$ = +25°C | $P_T$ (2) $T_C$ = +25°C | $P_T$ (2) $T_{SP}$ = +25°C | $R_{\theta JA}$ (3) | $R_{\theta JC}$ (3) | $R_{\theta JSP}$ (3) | $V_{CBO}$ | $V_{EBO}$ | $V_{CEO}$ | $I_C$ | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | W | °C/W | °C/W | °C/W | V dc | V dc | V dc | A dc | °C |
| 2N3439 | 0.8 | 5.0 | N/A | 175 | 30 | N/A | 450 | 7 | 350 | 1.0 | -65 to +200 |
| 2N3439L | 0.8 | 5.0 | N/A | 175 | 30 | N/A | 450 | 7 | 350 | 1.0 | |
| 2N3439UA | 0.8 | N/A | 2.0 | 175 | N/A | 70 | 450 | 7 | 350 | 1.0 | |
| 2N3439U4 | 0.8 | 5 | N/A | 175 | 8 | N/A | 450 | 7 | 350 | 1.0 | |
| 2N3440 | 0.8 | 5.0 | N/A | 175 | 30 | N/A | 300 | 7 | 250 | 1.0 | |
| 2N3440L | 0.8 | 5.0 | N/A | 175 | 30 | N/A | 300 | 7 | 250 | 1.0 | |
| 2N3440UA | 0.8 | N/A | 2.0 | 175 | N/A | 70 | 300 | 7 | 250 | 1.0 | |
| 2N3440U4 | 0.8 | 5 | N/A | 175 | 8 | N/A | 300 | 7 | 250 | 1.0 | |

(1) For derating, see figure 6.
(2) For derating, see figures 7, 8, and 9.
(3) For thermal impedance curves see figures 10, 11, 12, and 13.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact in formation can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .



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1.4 Primary electrical characteristics unless otherwise specified $T_A$ = +25°C.

| | $h_{FE2}$ (1)<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 2 mA dc | $h_{FE1}$ (1)<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 20 mA dc | $\|h_{fe}\|$<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 10 mA dc<br>f = 5 MHz | $C_{obo}$<br>$V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>100 kHz ≤ f ≤ 1 MHz | $V_{BE(sat)}$ (1)<br>$I_C$ = 50 mA dc<br>$I_B$ = 4 mA dc | $V_{CE(sat)}$<br>$I_C$ = 50 mA dc<br>$I_B$ = 4 mA dc |
|---|---|---|---|---|---|---|
| | | | | pF | V dc | V dc |
| Min | 30 | 40 | 3 | | | |
| Max | | 160 | 15 | 10 | 1.3 | 0.5 |

(1) Pulsed, (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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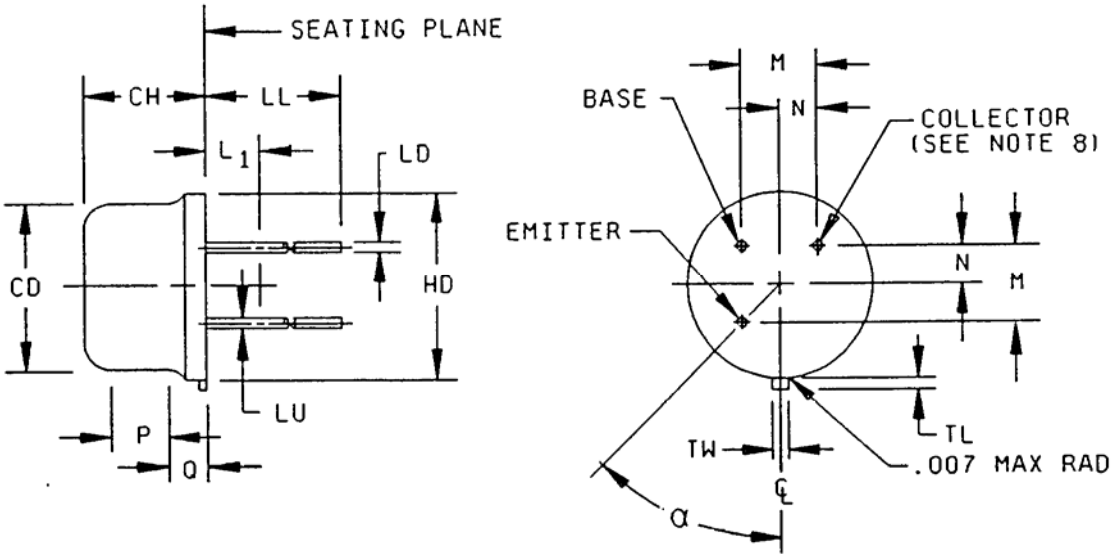



FIGURE 1. Physical dimensions (similar to TO-5 and TO-39).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LD | .016 | .021 | 0.41 | 0.53 | 3, 10 |
| LL | See notes 10, 12, and 13 | | | | |
| L1 | | .050 | | 1.27 | 11 |
| LU | .016 | .019 | 0.41 | 0.48 | 4, 10 |
| M | .1414 Nom | | 3.591 Nom | | 7 |
| N | .0707 Nom | | 1.796 Nom | | 7 |
| P | .100 | | 2.54 | | 5 |
| Q | | | | | 6 |
| TL | .029 | .045 | 0.74 | 1.14 | 9 |
| TW | .028 | .034 | 0.71 | 0.86 | |
| $\alpha$ | 45° | | 45° | | 7 |

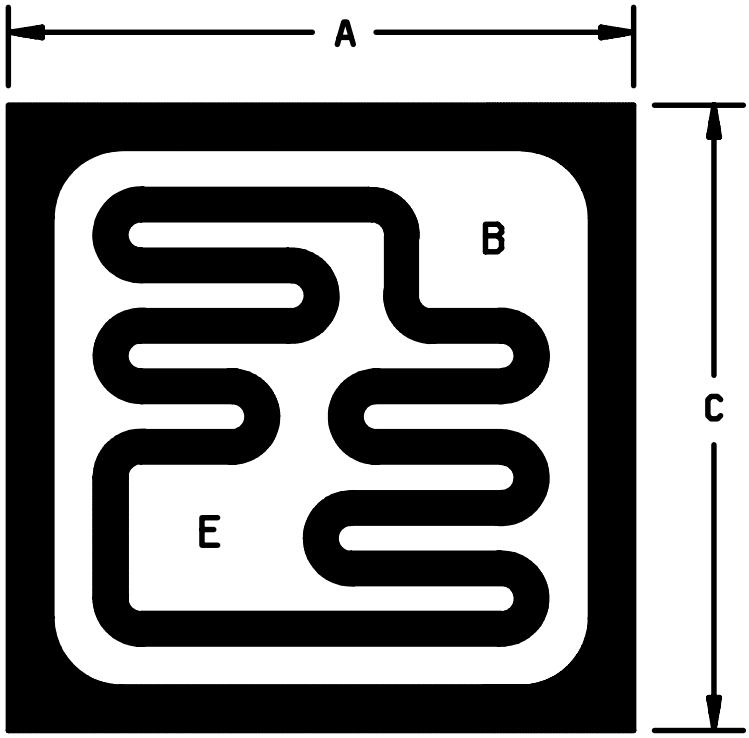
NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .50 inch (12.7 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 +.001, -.000 inch (1.37 +0.03, - 0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.4.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N3439 and 2N3440, L = .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum.
13. For transistor types 2N3439L and 2N3440L, L = 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.
14. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 1. Physical dimensions - Continued.



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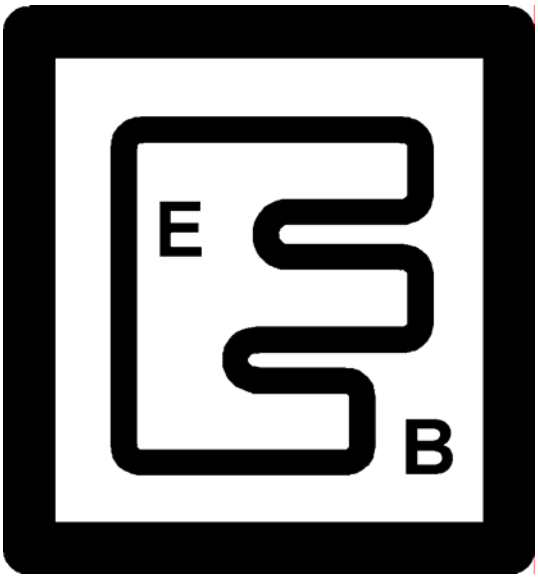

| Ltr | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | .038 | .044 | 0.97 | 1.12 |
| C | .038 | .044 | 0.97 | 1.12 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die thickness are, .006 inch (0.15 mm) to .012 inch (0.30 mm).
   Metallization, top: Al = 17,500 Å minimum, 20,000 Å nominal.
   Back: Au = 2,500 Å minimum, 3,000 Å nominal.
   Bonding pad: B = .004 inch (0.10 mm) by .005 inch (0.13 mm).
   E = .004 inch (0.10 mm) by .005 inch (0.13 mm).
4. Backside is collector.
5. Requirements in accordance with MIL-PRF-19500 are performed in a TO-5 package.
6. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 2. Physical dimensions JANHCA and JANKCA (die) A versions.



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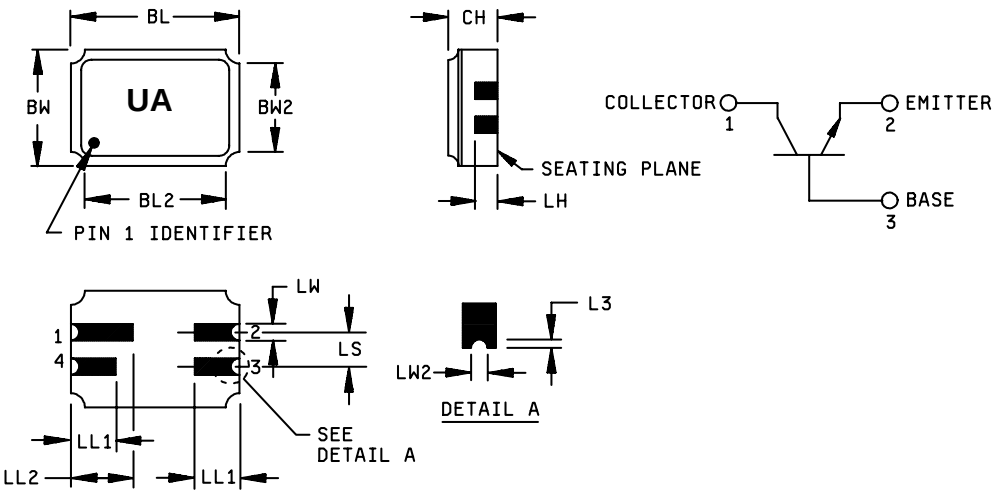

| | | |
|---|---|---|
| 1. | Chip size..... | .040 x .040 inch ±.002 inch (1.02 mm x 1.02 mm ±0.05 mm). |
| 2. | Chip thickness..... | .010 ± .0015 inch nominal (0.254 mm ±0.038 mm). |
| 3. | Top metal..... | Aluminum 30,000Å minimum, 33,000Å nominal. |
| 4. | Back metal..... | A. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅminimum,15kÅ/ 5kÅ/10kÅ/10kÅ nominal.<br>B. Gold 2,500Å minimum, 3000Å nominal. |
| 5. | Backside..... | Collector. |
| 6. | Bonding pad..... | B = .005 x .008 inch (0.127 mm x 0.203 mm).<br>E = .010 x .007 inch (0.254 mm x 0.178 mm). |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions JANHCB and JANKCB (die) B versions.



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| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BL | .215 | .225 | 5.46 | 5.71 | |
| BL2 | | .225 | | 5.71 | |
| BW | .145 | .155 | 3.68 | 3.93 | |
| BW2 | | .155 | | 3.93 | |
| CH | .061 | .075 | 1.55 | 1.90 | 3 |
| L3 | .003 | .007 | 0.08 | 0.18 | 5 |
| LH | .029 | .042 | 0.74 | 1.07 | |
| LL1 | .032 | .048 | 0.81 | 1.22 | |
| LL2 | .072 | .088 | 1.83 | 2.23 | |
| LS | .045 | .055 | 1.14 | 1.39 | |
| LW | .022 | .028 | 0.56 | 0.71 | |
| LW2 | .006 | .022 | 0.15 | 0.56 | 5 |

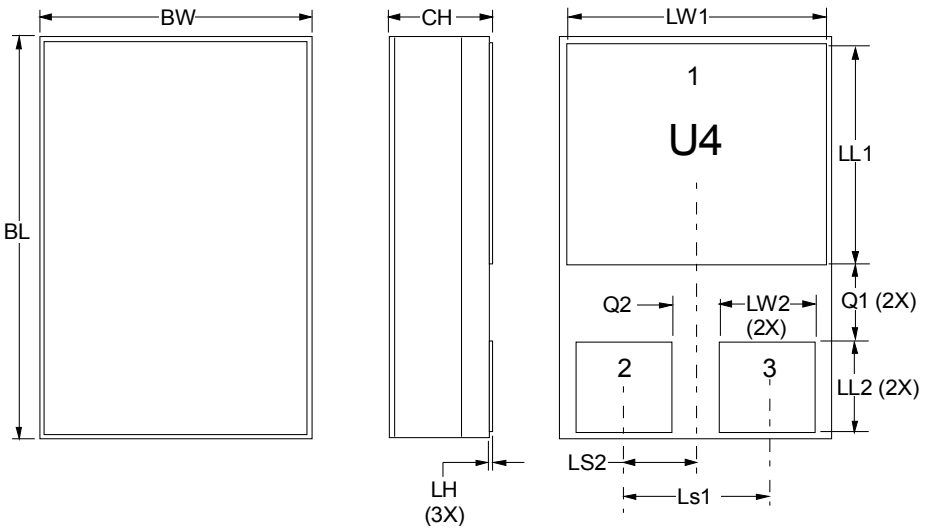
| Pin no. | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Transistor | Collector | Emitter | Base | N/C |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension "CH" controls the overall package thickness. When a window lid is used, dimension "CH" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions " LW2" minimum and "L3" minimum and the appropriate castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension " LW2" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.
6. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15mm) for solder dipped leadless chip carriers.
7. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions, surface mount (2N3439UA, 2N3440UA) version.



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .215 | .225 | 5.46 | 5.72 |
| BW | .145 | .155 | 3.68 | 3.94 |
| CH | .049 | .075 | 1.24 | 1.91 |
| LH | | .005 | | 0.127 |
| LW1 | .135 | .145 | 3.43 | 3.68 |
| LW2 | .047 | .057 | 1.19 | 1.45 |
| LL1 | .115 | .125 | 2.91 | 3.17 |
| LL2 | .045 | .055 | 1.14 | 1.40 |
| LS1 | .070 | .095 | 1.78 | 2.41 |
| LS2 | .038 | .048 | 0.965 | 1.21 |
| Q1 | .045 | .070 | 1.14 | 1.78 |
| Q2 | .025 | .035 | 0.635 | 0.88 |
| 1 | Collector | | | |
| 2 | Base | | | |
| 3 | Emitter | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1 is collector.
4. Terminal 2 is base.
5. Terminal 3 is emitter.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions, surface mount (2N3439U4, 2N3440U4) version (U4).



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

| | |
|---|---|
| $R_{\theta JA}$ | Thermal resistance junction to ambient. |
| $R_{\theta JC}$ | Thermal resistance junction to case. |
| $R_{\theta JSP}$ | Thermal resistance junction to solder pads. |
| UA | Surface mount case outlines (see figure 4). |
| TRB | Technical review board. |
| $T_{SP}$ | Temperature of solder pads. |

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (similar to TO- 5 and TO-39), figure 2 (JANHCA and JANKCA, (A versions)), figure 3 (JANHCB and JANKCB (B versions)), figure 4 (2N3439UA and 2N3440UA surface mount, and figure 5 (2N3439U4 and 2N3440U4)).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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* 4.2.1.1 Group E thermal impedance. Each supplier shall submit a thermal impedance ($Z_{\theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal impedance. Thermal impedance curves (from $Z_{\theta JX}$ test pulse time to $R_{\theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal impedance screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\theta JX}$ shall also be submitted. The proposed specification maximum thermal impedance curve shall be submitted. The qualifying activity may approve a different $Z_{\theta JX}$ limit not to exceed the specification's thermal curve for conformance inspection end-point measurements as applicable**.** Equivalent data, procedures, or statistical process control plans may be used for part, or all, of the above requirements. The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and table I, subgroup 2. The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.

* 4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3) | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3) |
| 7 | Hermetic seal (optional) | Optional |
| 9 | $I_{CBO1}$ and $h_{FE1}$ | Not applicable |
| 10 | 48 hours minimum | 48 hours minimum |
| 11 | $I_{CBO1}$; $h_{FE1}$; $\Delta I_{CB01}$ = 100 percent of initial value or 0.5 μA dc, whichever is greater; $\Delta h_{FE1}$ = ± 15 percent of initial value. | $I_{CBO1}$ and $h_{FE1}$ |
| 12 | See 4.3.2 | See 4.3.2 |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{CB01}$ = 100 percent of initial value or 200 nA dc, whichever is greater; $\Delta h_{FE1}$ = ± 15 percent of initial value. | Subgroup 2 of table I herein; $\Delta I_{CB01}$ = 100 percent of initial value or 200 nA dc, whichever is greater; $\Delta h_{FE1}$ = ± 15 percent of initial value. |
| 14 | Required | Required |

(1) Thermal impedance, see 4.3.3. Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to ensure the assembled chips will meet the requirements of table I, subgroup 2.



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* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in VIa (JANS) of 4.4.2.1 herein. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.2 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.2 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CB}$ = 10 V dc, 2,000 cycles. |
| B5 | 1027 | $V_{CB}$ = 10 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3).<br>(NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)<br><br>Option 1: 96 hrs minimum, sample size in accordance with table VIa of MIL-PRF-19500, adjust $T_A$ or $P_D$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hrs minimum, sample size = 45, c = 0; adjust $T_A$ or $P_D$ to achieve $T_J$ = +225°C minimum. |
| B6 | 3131 | $R_{\theta JA}$ for TO-5 and TO-39, $R_{\theta JC}$ for U4 only (see 1.3 and 4.3.3). |



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* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours, $V_{CB}$ = 10 dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, VCB = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV, samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.2 herein.

* 4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E, except the UA and U4 packages. |
| C5 | 3131 | $R_{\theta JA}$ for TO-5 and TO-39, $R_{\theta JC}$ for U4 only, as applicable (see 1.3 and see 4.3.3) and in accordance with thermal impedance curves. |
| C6 | 1026 | 1,000 hours at $V_{CB}$ = 10 V dc; power shall be applied to achieve $T_J$ = +150°C minimum and a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |



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* 4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Except the UA and U4 packages. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.3) and in accordance with thermal impedance curves. |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with the steps of 4.5.3.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Delta requirements. Delta requirements shall be as specified below: (1) (2) (3) (4) (5)

| Step | Inspection | MIL-STD-750 | | Symbol | Unit |
|---|---|---|---|---|---|
| | | Method | Conditions | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D,<br>$V_{CB}$ = 360 V dc for 2N3439, 2N3439L, 2N3439UA, 2N3439U4<br>$V_{CB}$ = 250 V dc for 2N3440, 2N3440L, 2N3440UA, 2N3439U4 | $\Delta I_{CB01}$ | 100 percent of initial value or 200 nA dc, whichever is greater. |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc;<br>$I_C$ = 20 mA dc;<br>pulsed see 4.5.1 | $\Delta h_{FE1}$ | ±15 percent change from initial reading. |

(1) Devices which exceed the table I limits for this test shall not be accepted.
(2) The delta measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:
  a. B4, see 4.5.3 herein, steps 1 and 2.
  b. B5, see 4.5.3 herein, steps 1 and 2.
(3) The delta measurements for 4.4.2.2 herein are as follows: Step 1, see 4.5.3 herein, steps 1 and 2.
(4) The delta measurements for appendix E, table VII (JANS) of MIL-PRF-19500 are as follows: C6, see 4.5.3 herein, steps 1 and 2.
(5) The delta measurements for appendix E, table IX of MIL-PRF-19500 are as follows:
  a. E1, see 4.5.3, steps 1 and 2.
  b. E2, see 4.5.3, steps 1 and 2.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical 3/ examination | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to 3/ 4/ 5/ solvent | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Heremetic seal 4/ | 1071 | n = 22 devices, c = 0 | | | | |
| Fine leak<br>Gross leak | | | | | | |
| Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hrs or $T_A$ = +300°C at t = 2 hrs, n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) 4/ | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.3 | $Z_{\theta JX}$ | | | °C/W |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 7 V dc | $I_{EBO1}$ | | 10 | µA dc |
| Collector to emitter cutoff | 3041 | Bias condition D | $I_{CEO}$ | | | |
| 2N3439, 2N3439L, 2N3439UA, 2N3439U4 | | $V_{CE}$ = 300 V dc | | | 2 | µA dc |
| 2N3440, 2N3440L, 2N3440UA, 2N3440U4 | | $V_{CE}$ = 200 V dc | | | 2 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = -1.5 V dc | $I_{CEX}$ | | | |
| 2N3439, 2N3439L, 2N3439UA, 2N3439U4 | | $V_{CE}$ = 450 V dc | | | 5 | µA dc |
| 2N3440, 2N3440L, 2N3440UA, 2N3440U4 | | $V_{CE}$ = 300 V dc | | | 5 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO1}$ | | | |
| 2N3439, 2N3439L, 2N3439UA, 2N3439U4 | | $V_{CB}$ = 360 V dc | | | 2 | µA dc |
| 2N3440, 2N3440L, 2N3440UA, 2N3440U4 | | $V_{CB}$ = 250 V dc | | | 2 | µA dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO2}$ | | | |
| 2N3439, 2N3439L, 2N3439UA, 2N3439U4 | | $V_{CB}$ = 450 V dc | | | 5 | µA dc |
| 2N3440, 2N3440L, 2N3440UA, 2N3440U4 | | $V_{CB}$ = 300 V dc | | | 5 | µA dc |
| Base emitter voltage (nonsaturated) | 3066 | Test condition A, pulsed (see 4.5.1), $I_C$ = 50 mA dc, $I_B$ = 4 mA dc | $V_{BE(sat)}$ | | 1.3 | V dc |
| Collector to emitter voltage (saturated) | 3071 | Pulsed (see 4.5.1), $I_C$ = 50 mA dc, $I_B$ = 4 mA dc | $V_{CE(sat)}$ | | 0.5 | V dc |
| Forward-current transfer ratio | 3076 | Pulsed (see 4.5.1), $V_{CE}$ = 10 V dc, $I_C$ = 20 mA | $h_{FE1}$ | 40 | 160 | |
| Forward-current transfer ratio | 3076 | Pulsed (see 4.5.1), $V_{CE}$ = 10 V dc, $I_C$ = 2 mA | $h_{FE2}$ | 30 | | |
| Forward-current transfer ratio | 3076 | Pulsed (see 4.5.1), $V_{CE}$ = 10 V dc, $I_C$ = 0.2 mA | $h_{FE3}$ | 10 | | |
| Subgroup 3 | | $T_A$ = +150°C | | | | |
| High temperature operation: | | | | | | |
| Collector to emitter cutoff current | 3036 | Bias condition D | $I_{CB03}$ | | | |
| 2N3439, 2N3439L, 2N3439UA, 2N3439U4 | | $V_{CB}$ = 360 V dc | | | 6 | µA dc |
| 2N3440, 2N3440L, 2N3440UA, 2N3440U4 | | $V_{CB}$ = 250 V dc | | | 6 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 20 mA dc, pulsed (see 4.5.1) | $h_{FE4}$ | 15 | | |
| Subgroup 4 | | | | | | |
| Pulse response: | 3251 | Test condition A | | | | |
| Turn-on time | | $V_{CC}$ = 200 V dc, $I_C$ = 20 mA dc, $I_{B1}$ = 2 mA dc, see figure 14 | $t_{on}$ | | 1 | µs |
| Turn-off time | | $V_{CC}$ = 200 V dc, $I_C$ = 20 mA dc, $I_{B1}$ = -$I_{B2}$ = 2 mA dc, see figure 14 | $t_{off}$ | | 10 | µs |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Magnitude of common- emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc, $I_C$ = 10 mA dc, f = 5 MHz | $\|h_{fe}\|$ | 3 | 15 | |
| Open capacitance input open circuited | 3236 | $V_{CB}$ = 10 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MH | $C_{obo}$ | | 10 | pF |
| Small-signal short- circuit forward-current transfer ratio | 3206 | $V_{CE}$ = 10 V dc, $I_C$ = 5 mA, f = 1 kHz | $h_{fe}$ | 25 | | |
| Input capacitance (output open circuited) | 3240 | $V_{CB}$ = 5 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 75 | pF |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | (See figure 15) $T_C$ = +25°C, 1 cycle, t = 1.0 s. | | | | |
| Test 1 | | $V_{CE}$ = 5 V dc, $I_C$ = 1 A dc | | | | |
| Test 2 | | | | | | |
| Only 2N3439, 2N3439L, 2N3439UA | | $V_{CE}$ = 350 V dc, $I_C$ = 14 mA dc | | | | |
| Test 3 | | | | | | |
| Only 2N3440, 2N3440L, 2N3440UA | | $V_{CE}$ = 250 V dc, $I_C$ = 20 mA dc | | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.3 herein. | | | | |
| Breakdown voltage, collector to emitter | 3011 | $I_C$ = 10 mA, $R_{BB1}$ = 470 ohms $V_{BB1}$ = 6 V, L = 25 mH minimum f = 30 to 60 Hz | $V_{BR(CEO)}$ | | | V dc |
| 2N3439, 2N3439L, 2N3439UA<br>2N3440, 2N3440L, 2N3440UA | | | | 350<br>250 | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS.
5/ Not required for laser marked devices.



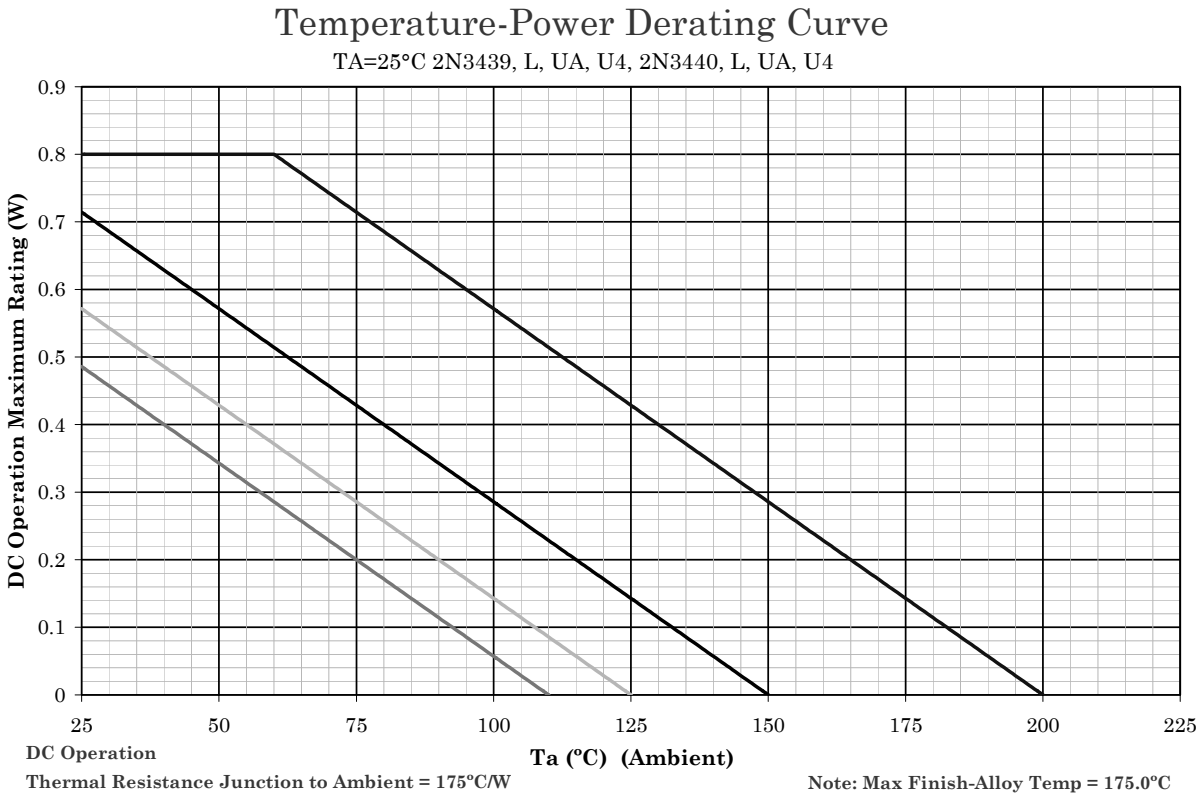
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* TABLE II. Group E inspection (all quality levels) - for qualification only or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.2. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB} \geq 10$ V dc, 6,000 cycles | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.2. | |
| Subgroup 4 | | | |
| Thermal resistance | 3131 | $R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). | 15 devices,<br>c = 0 |
| Thermal impedance curves | | See 4.2.1.1. | sample size<br>N/A |
| Subgroup 5 | | | 15 devices,<br>c = 0 |
| Barometric pressure (reduced) | 1001 | Normal mounting pressure = 8 mm Hg<br>±2 mm Hg for 60 s (minimum) | |
| Electrical measurements | | See table I, subgroup 2. | |
| Subgroups 6 | | | 3 devices<br>c = 0 |
| ESD (electrocstatic discharge) | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition A for devices ≥ 400 V,<br>condition B for devices < 400 V. | |



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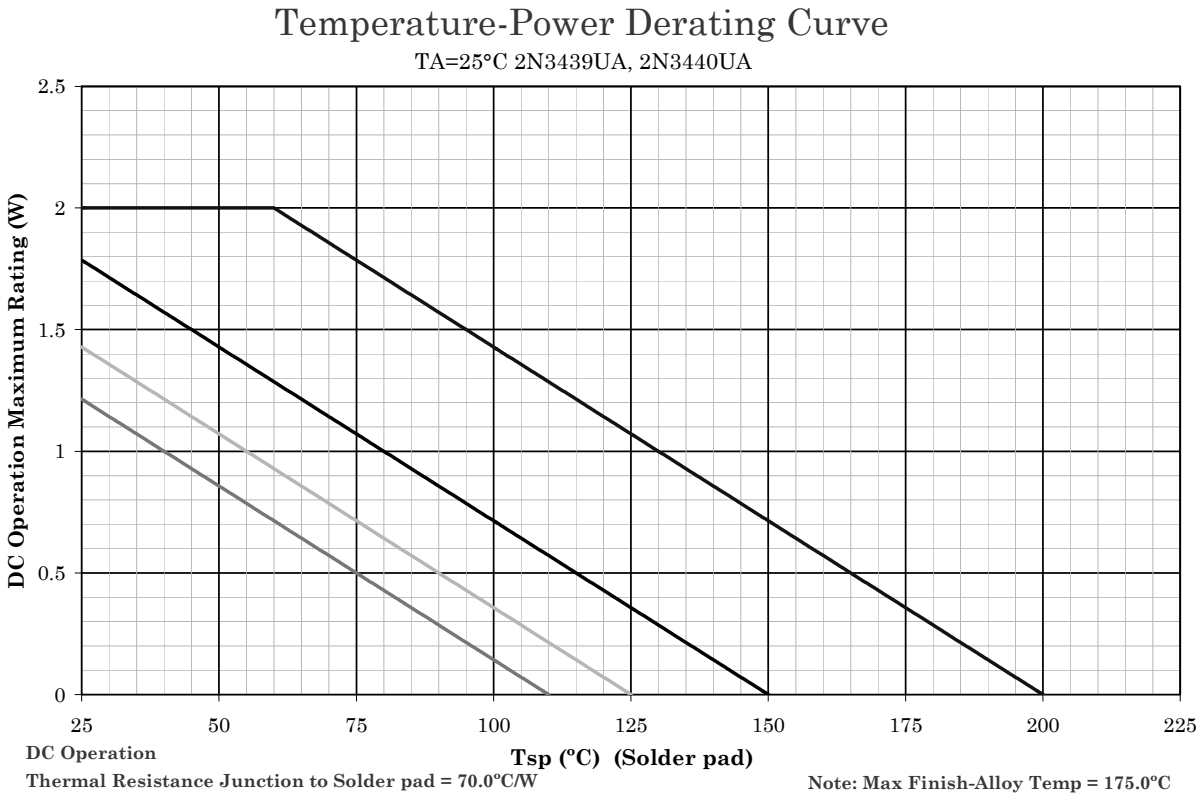

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 6. Temperature-power derating for all types $R_{\theta JA}$ (TO-5, TO-39, UA and U4).



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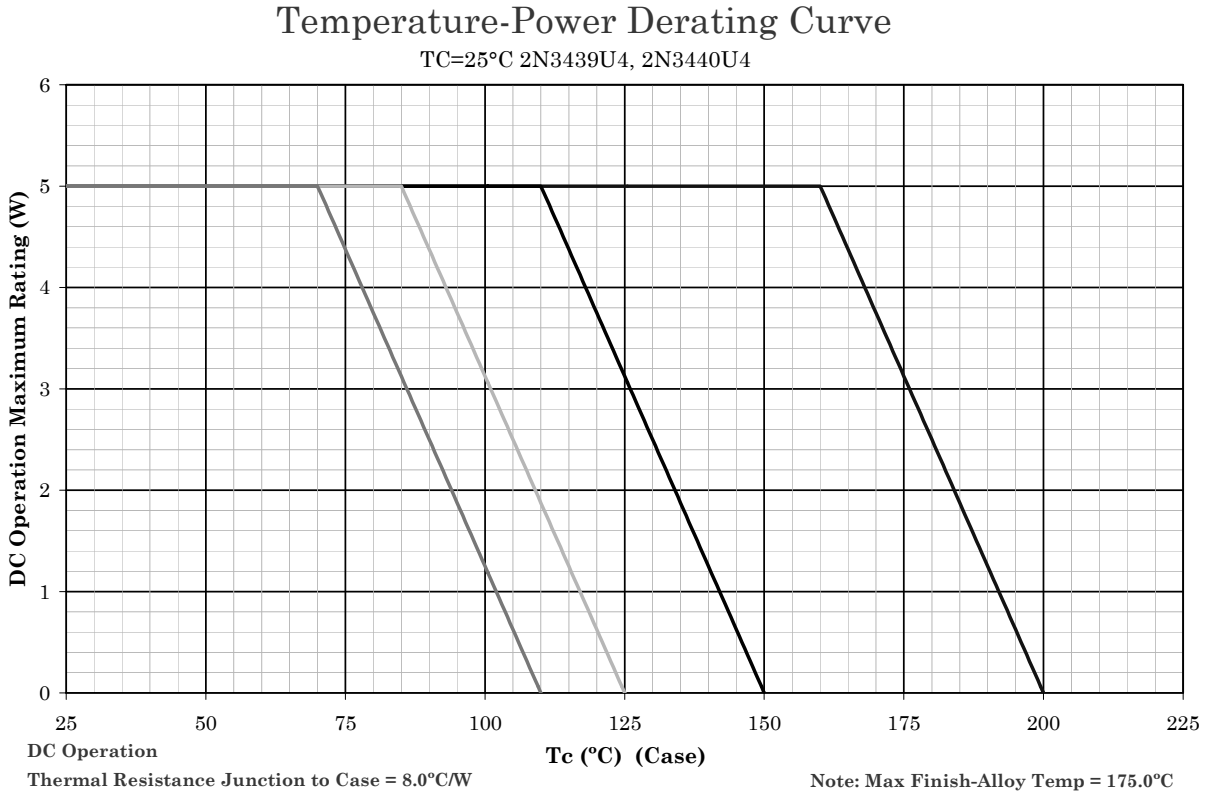

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 7. Temperature-power derating for 2N3439UA and 2N3440UA.



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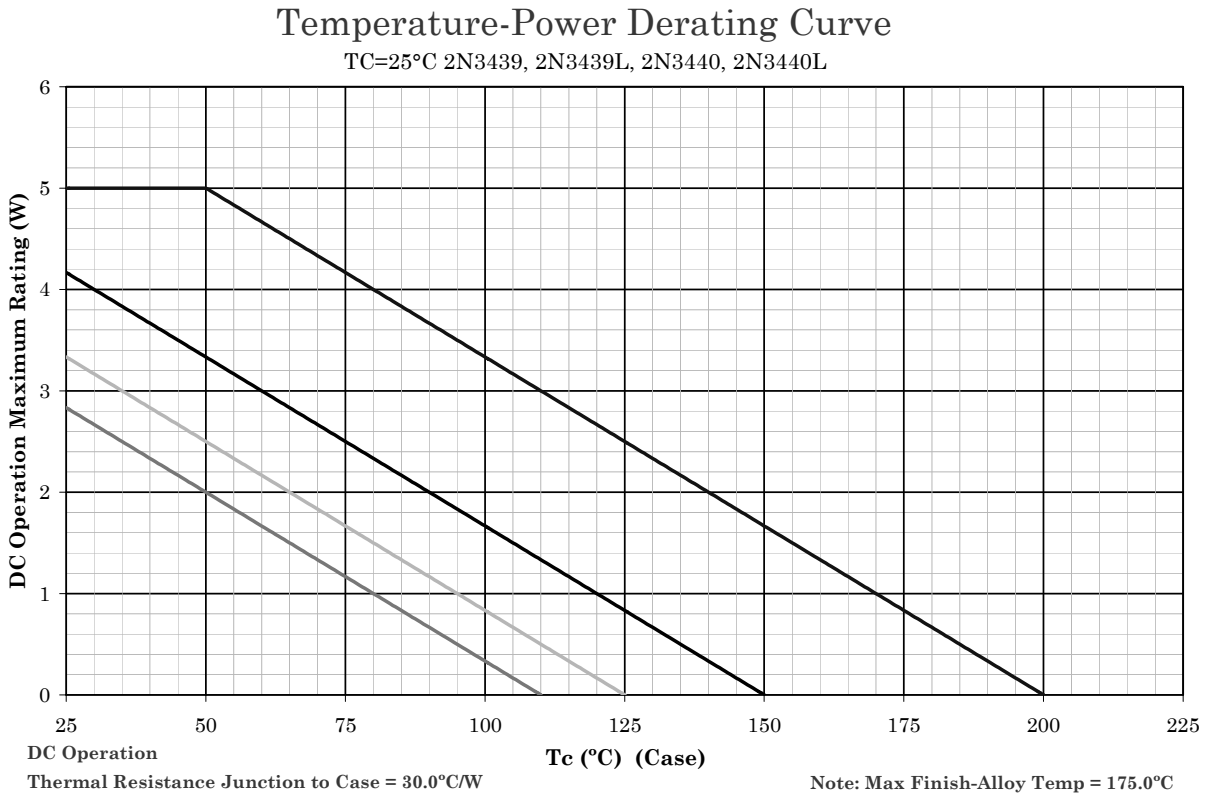

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 8. Temperature-power derating for 2N3439U4 and 2N3440U4.



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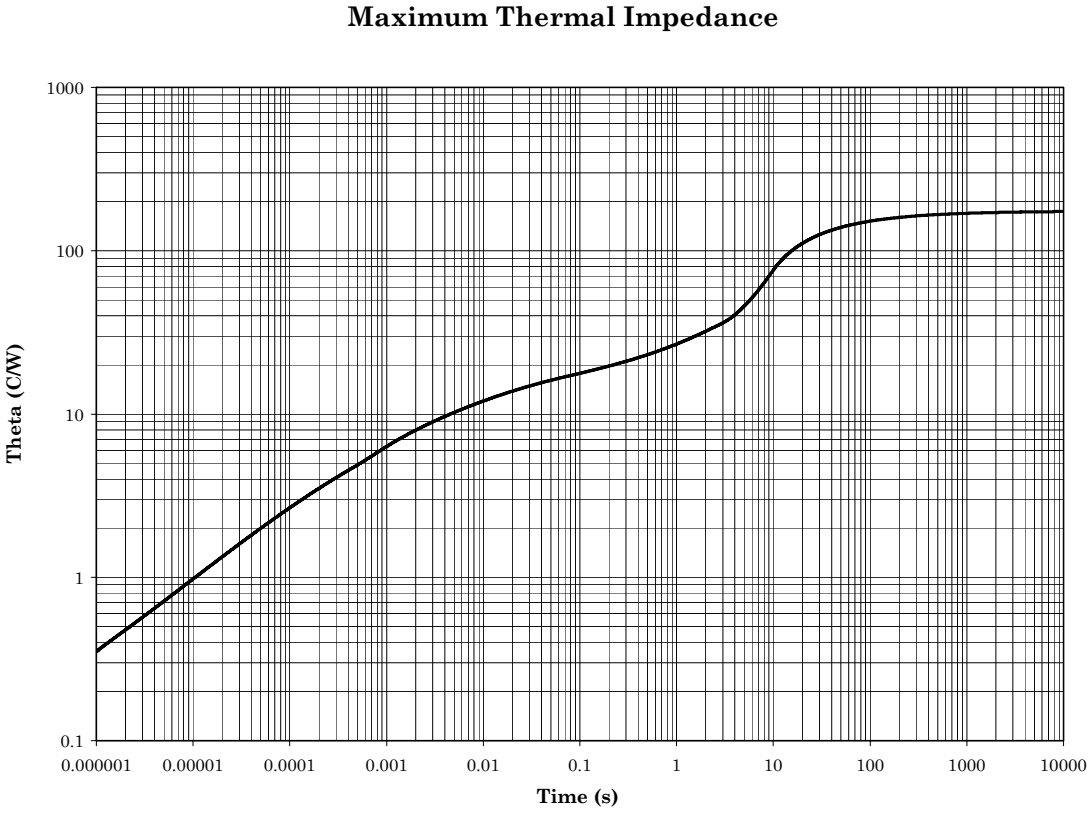

NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit $T_J$ in their application.

* FIGURE 9. Temperature-power derating for 2N3439, 2N3439L, 2N3440, and 2N3440L.



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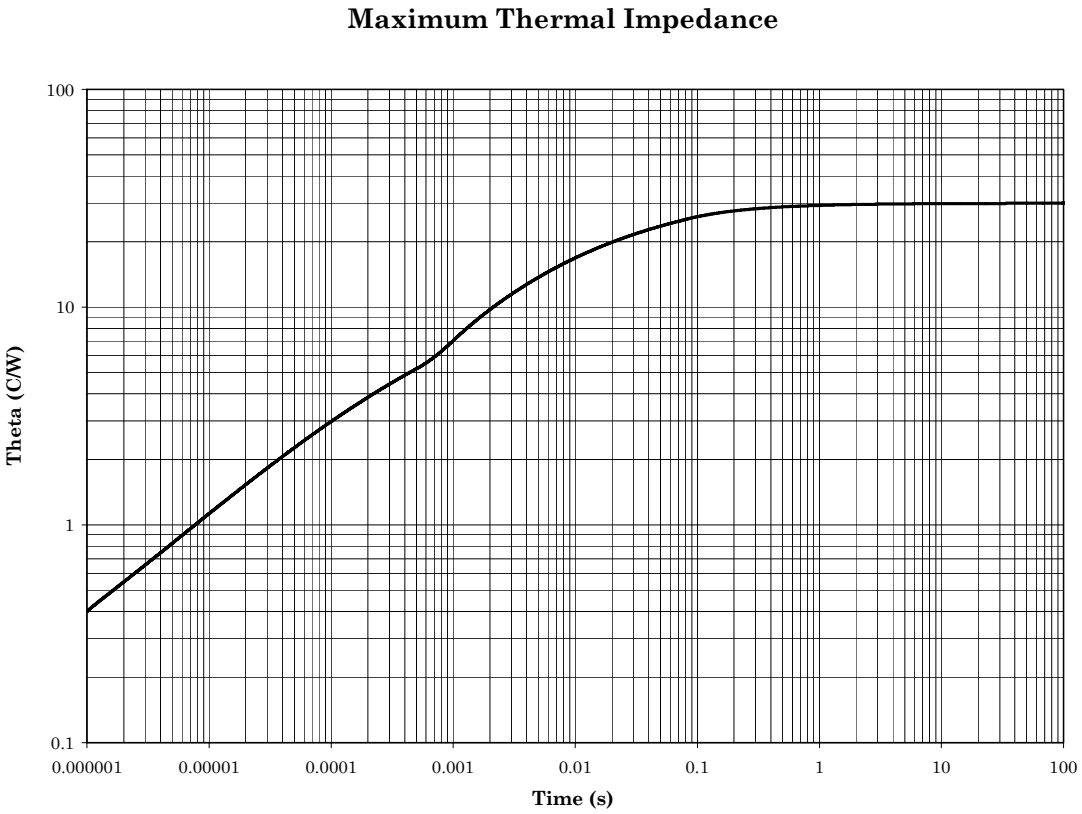



$T_A$ = +25°C, $P_T$ = 0.8W, thermal resistance $R_{\theta JA}$ = 175°C/W.

FIGURE 10. Thermal impedance graph ($R_{\theta JA}$) for 2N3439, 2N3439L, 2N3440, and 2N3440L (TO-5, TO-39, U4, and UA).



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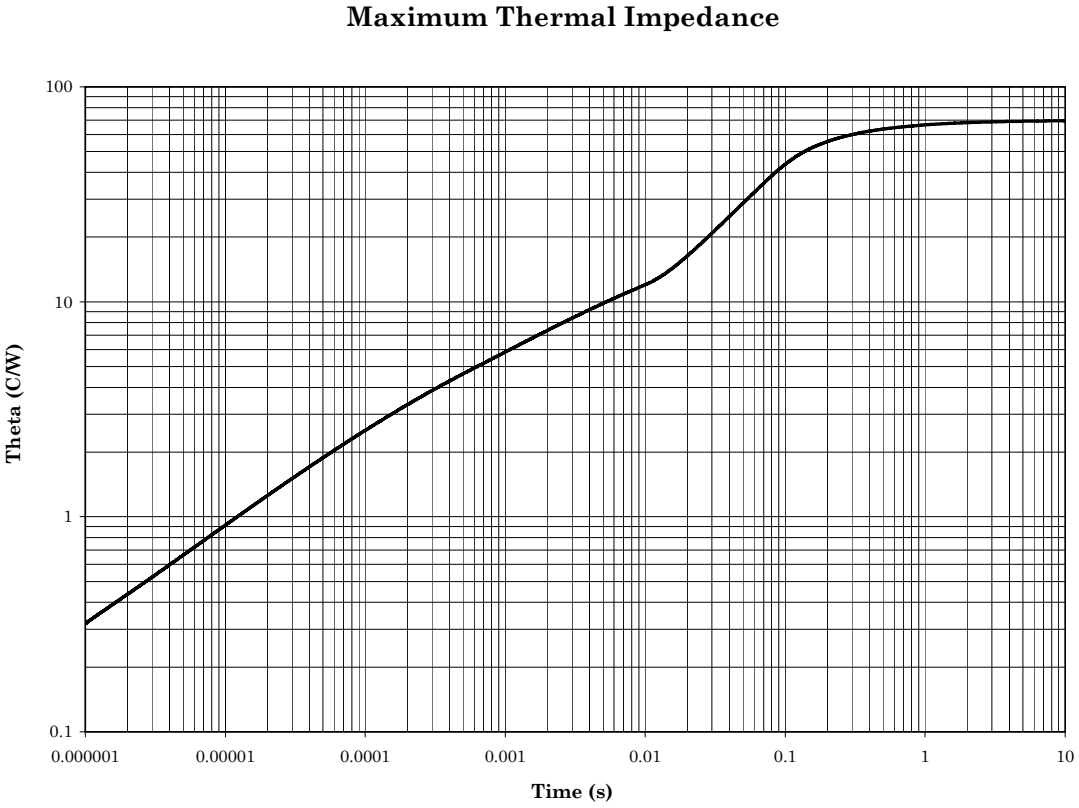



$T_C$ = +25°C, $P_T$ = 5.0W, thermal resistance $R_{\theta JC}$ = 30°C/W, steel.

FIGURE 11. Thermal impedance graph ($R_{\theta JC}$) for 2N3439, 2N3439L, 2N3440, and 2N3440L (TO-5 and TO-39).



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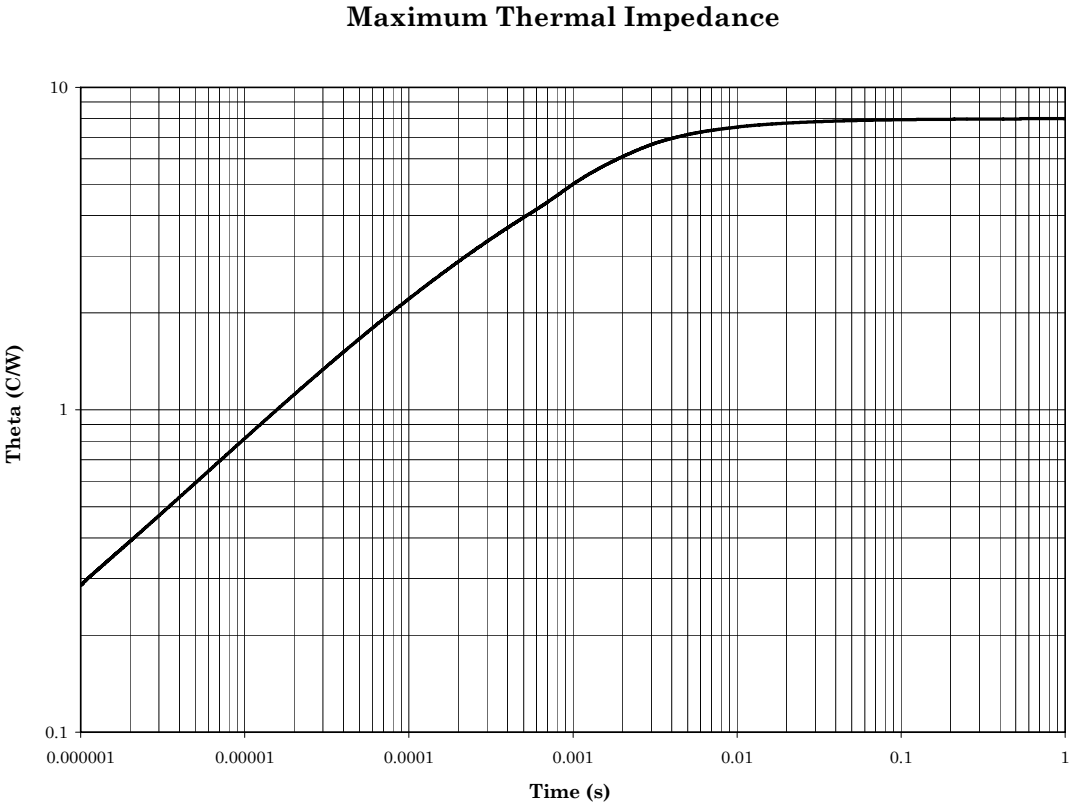



$T_C$ = +25°C, thermal resistance $R_{\theta JSP}$ = 70°C/W, Pdiss = 2W.

FIGURE 12. Thermal impedance graph ($R_{\theta JSP}$) for 2N3439, 2N3439L, 2N3439UA, 2N3440, 2N3440L, and 2N3440UA.



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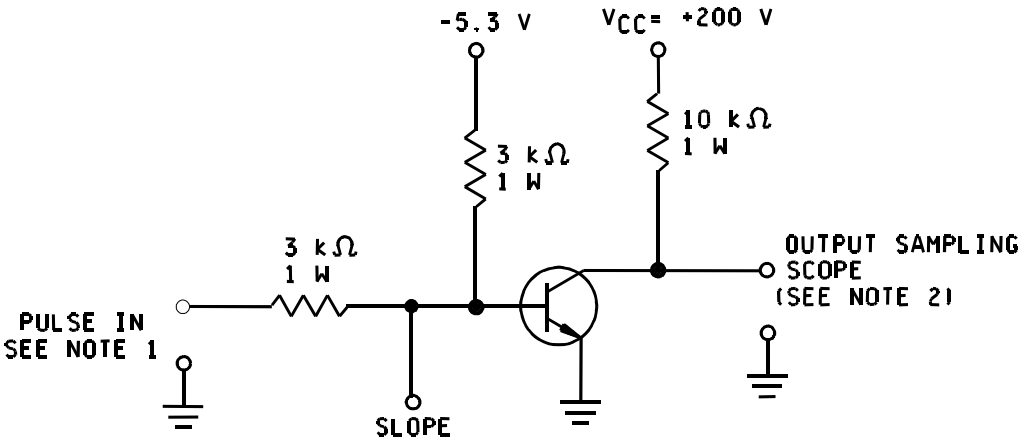


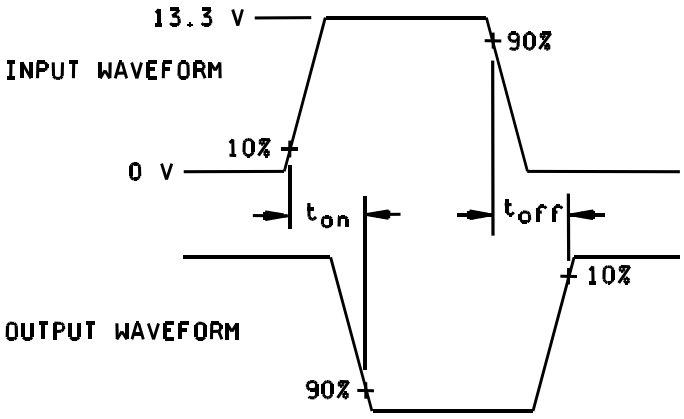

Solder mounted to copper heatsink at $T_C$ = +25C. Thermal resistance $R_{\theta JC}$ = 8.0 C/W

FIGURE 13. Thermal impedance graph ($R_{\theta JC}$) for 2N3439U4, and 2N3440U4 (U4).



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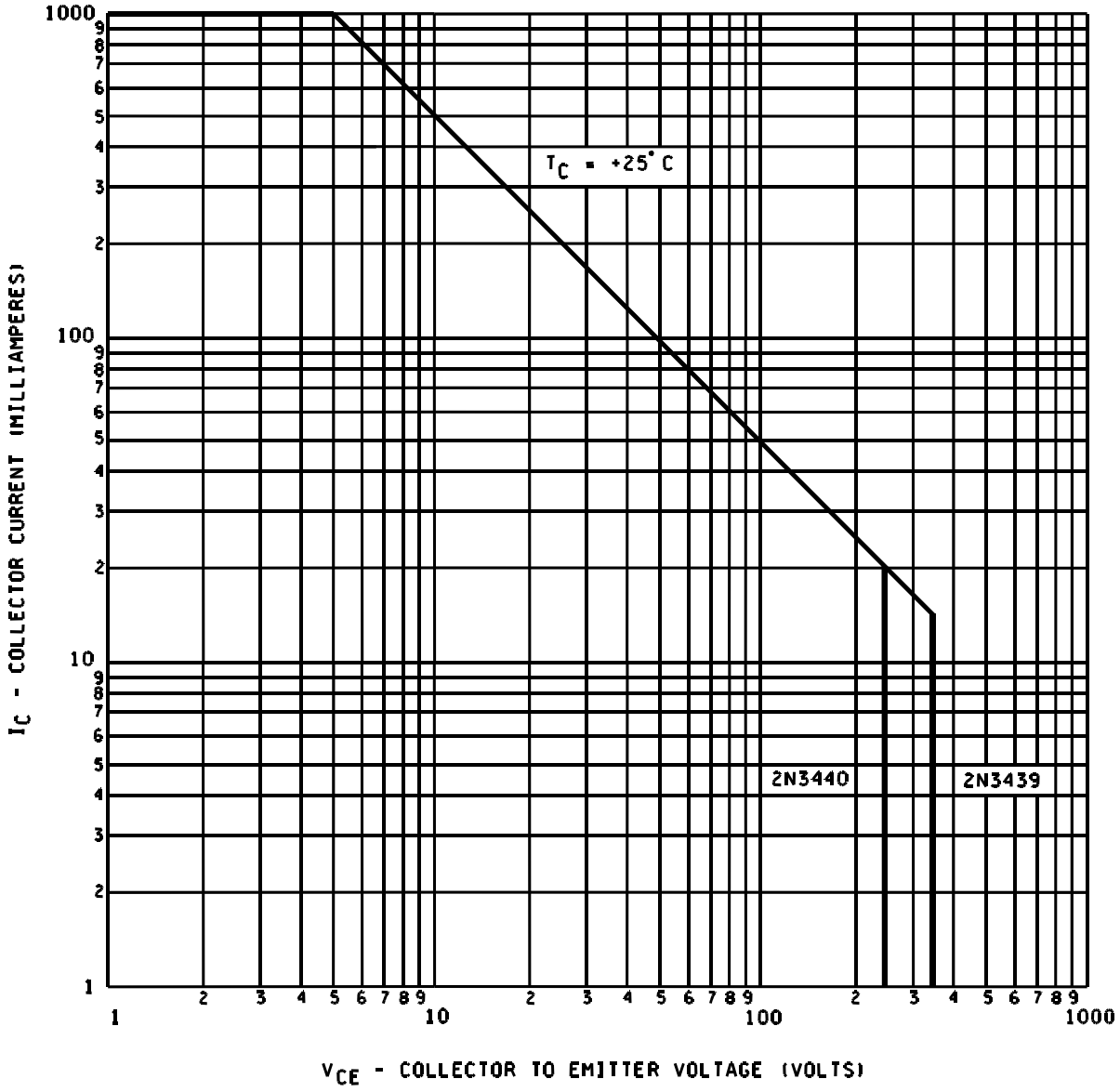

NOTES:

1. The rise time ($t_r$) and fall time ($t_f$) of applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, generator source impedance shall be 50 ohms, pulse width = 20 μs.
2. Output sampling oscilloscope: $Z_{IN}$ ≥ 100 kΩ, $C_{IN}$ ≤ 50 pF, and rise time ≤ 1.0 μs.

FIGURE 14. Pulse response test circuit.



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NOTE: Also applies to the corresponding "L" and "UA" suffix devices.

FIGURE 15. Maximum safe operating graph (continuous dc).



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3439) will be identified on the QML.

| Die ordering information | | |
|---|---|---|
| PIN | Manufacturer | |
| | 33178 | 43611 |
| 2N3439 | JANHCA2N3439<br>JANKCA2N3439 | JANHCB2N3439<br>JANKCB2N3439 |
| 2N3440 | JANHCA2N3440<br>JANKCA2N3440 | JANHCB2N3440<br>JANKCB2N3440 |



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2005-033)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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